UNIVERSITY MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE & MATHEMATICS

SOFTWARE ARCHITECTURE
CSE3433

INDIVIDUAL LOGBOOK

GROUP NO. 16

SERVICE-ORIENTED SMART CAMPUS SUPPORT SYSTEM

PREPARED BY:
AFIQ FAHIM BIN MOHD RIDZUAN

PREPARED FOR:
DR. MOHAMAD NOR HASAN

[BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOURS]
SEMESTER 2 2025/2026

1.0 Team Information

Name	Matric No	Role
AFIQ FAHIM BIN MOHD RIDZUAN	S76096	SYSTEM ARCHITECT

2.0 Weekly Progress Report

Week 4

Item	**Description**
Week	Week 4 (2 - 8 March 2026)
Weekly Activities	The team discussed the project concept, identified the problem statement and analysed the main requirements.
Individual Contributions	As the System Architect, I analysed the overall system flow and identifying the main components required for the application.
Challenges	The main challenge was determining how to divide the system into separate components while ensuring that each component had a clear responsibility and could work together efficiently.
Action Plans	Study suitable architecture styles and prepare the initial system structure for the architecture design phase.

Week 5

Item	Description
Week	Week 5 (9 - 15 March 2026)
Weekly Activities	Completed the Project Proposal document.
Individual Contributions	I contributed to the technical sections of the proposal by explaining the planned system structure and suggesting the use of MVC architecture. I also reviewed whether the proposed features could be supported by the planned design.
Challenges	The challenge was ensuring that the selected architecture was suitable for the project size while still allowing future expansion of the system.
Action Plans	Begin designing the architecture components and prepare diagrams required for Deliverable 2.

Week 6

Item	Description
Week	Week 6 (16 - 22 March 2026)
Weekly Activities	The team started preparing the Architecture Design document by identifying system components and defining interactions between modules.
Individual Contributions	I designed the initial MVC structure by separating the system into Model, View and Controller layers.
Challenges	Designing the communication flow between components was hard because multiple modules needed to access shared information from the database.
Action Plans	Complete the architecture diagrams and ensure that the component relationships match the planned implementation.

Week 7

Item	Description
Week	Week 7 (23 - 29 March 2026)
Weekly Activities	The team completed the Architecture Design document including the system overview, requirements, architecture explanation and required diagrams.
Individual Contributions	I prepared and reviewed the architecture diagrams including the component structure and interaction flow.
Challenges	The main challenge was maintaining consistency between the diagrams and the actual implementation plan because changes to one component could affect the overall design.
Action Plans	Start implementing the system based on the finalized architecture and assist team members during development.

Week 8

Item	Description
Week	Week 8 (30 March - 5 April 2026)
Weekly Activities	Development started and the team began implementing different system modules based on the architecture design.
Individual Contributions	I assisted with backend implementation by working on the authentication workflow and reviewing the structure of the LoginServlet.
Challenges	The main challenge was integrating different layers because frontend JSP pages, Servlets and backend logic needed to communicate correctly.
Action Plans	Continue improving authentication functionality and assist with integration between modules.

Week 9

Item	Description
Week	Week 9 (6 - 12 April 2026)
Weekly Activities	The team prepared and demonstrated the current system progress during Project Observation 1.
Individual Contributions	I presented the technical structure of the system and explained how the authentication module and MVC components worked together. I also helped test the existing features before the demonstration.
Challenges	Some modules were still incomplete, making it necessary to demonstrate only the completed functions while explaining future development plans.
Action Plans	Complete the remaining modules and improve integration before prototype submission.

Week 10

Item	Description
Week	Week 10 (13 - 19 April 2026)
Weekly Activities	The team continued development and integration testing before prototype submission.
Individual Contributions	I worked on role-based access control by reviewing the home page logic and ensuring different users received different system functions based on their roles. I also assisted in testing report viewing and navigation between modules.
Challenges	Handling different user roles was challenging because each role required different permissions and available functions.
Action Plans	Finalize prototype features and prepare evidence for Deliverable 3.

Week 11

Item	Description
Week	Week 11 (20 - 26 April 2026)
Weekly Activities	The team prepared the prototype progress submission with a working system demonstration, screenshots and repository evidence.
Individual Contributions	I verified that the implemented modules followed the designed architecture. I assisted in testing the login function, report viewing process and system navigation to ensure smooth interaction between components.
Challenges	The challenge was identifying integration issues because individual modules worked separately but sometimes failed when connected.
Action Plans	Perform further integration testing and prepare improvements based on feedback.

Week 12

Item	Description
Week	Week 12 (27 April – 3 May 2026)
Weekly Activities	The team conducted Project Observation 2 and demonstrated additional completed system functions.
Individual Contributions	I worked on integration improvements and assisted in resolving issues between modules. I reviewed the task acceptance workflow and ensured that updates between components were working correctly.
Challenges	The main challenge was fixing remaining issues while preparing the final report at the same time.
Action Plans	Complete final testing, fix remaining issues and support final documentation.

Week 13

Item	Description
Week	Week 13 (4 – 10 May 2026)
Weekly Activities	The team completed the final report and individual logbooks before the final demonstration.
Individual Contributions	I documented the architecture implementation, including MVC structure, component communication and technical decisions. I also

	reviewed the final system to ensure that implementation matched the original architecture design.
Challenges	The challenge was ensuring the final documentation accurately represented the implemented system and the contributions of each team member.
Action Plans	Prepare for the final demonstration and perform final system checks.

ChatGPT

3.0 GenAI Usage Record

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	ChatGPT	Understand software architecture concepts	"What are the fundamental principles of software architecture?"	Suggested possible entities such as users, disaster reports, volunteers, resources, tasks and notifications.	Used the suggestions as a reference when identifying the initial database entities for the system.
	ChatGPT	Explore architecture patterns	"What architecture patterns are commonly used in software development?"	Explained one-to-one, one-to-many and many-to-many relationships with examples.	Used the explanation when discussing possible relationships between users, reports and tasks.
Week 4	ChatGPT	Architecture selection support	"What architecture style is suitable for a Java Servlet and JSP based system?"	Explained that MVC architecture is suitable because it separates interface,	Used the information as a reference when selecting MVC architecture for the project proposal.

				business logic and data handling.	
Week 5	ChatGPT	Requirement analysis	"How can I identify functional and non-functional requirements for a software system?"	Provided examples of system requirements including security, usability and performance requirements.	Used the response to review whether our proposed requirements covered both functional and quality aspects.
	ChatGPT	Learn MVC principles	"Explain MVC architecture in Java web applications with examples."	Explained the roles of Model, View and Controller and how requests flow between components.	Used the explanation when designing the system structure and deciding where JSP pages, Servlets and Java classes belong.
Week 6	ChatGPT	Component identification	"How can I divide my system into separate components based on responsibilities?"	Suggested separating components according to features and responsibilities.	Used the suggestions to identify modules such as authentication, reporting, resources, tasks and notifications.
	ChatGPT	Architecture diagram guidance	"What should be included in a component architecture diagram?"	Explained how to represent system components and their relationships.	Used the explanation as a reference when preparing the Component Architecture Diagram.

Week 8	ChatGPT	Interaction flow design	"How do I design an interaction diagram for a web application?"	Explained how users, interfaces, controllers, services and databases communicate.	Used the explanation to review the communication flow between JSP pages, Servlets and database operations.
Week 9	ChatGPT	Servlet development support	"How does HttpServlet process GET and POST requests in Java?"	Explained request handling methods, parameters and response handling.	Used the explanation when reviewing Servlet implementation and ensuring correct request processing.
Week 10	ChatGPT	Debugging servlet issues	"Why is my servlet unable to retrieve form parameters from JSP?"	Suggested checking form method, input names, servlet mapping and request parameters.	Used the troubleshooting steps to identify possible issues between JSP forms and Servlets
	ChatGPT	Session management support	"How can I maintain user login sessions in JSP and Servlet?"	Explained the use of HttpSession for storing user information.	Used the explanation when reviewing the login process and session handling in the system.
Week 11	ChatGPT	Project presentation preparation	"How should I explain software architecture during a project presentation?"	Suggested explaining system flow, components, responsibilities and design decisions.	Used the structure to prepare explanations for Project Observation 1.

Week 12	ChatGPT	Role-based access design	"How can a JSP page display different features based on user roles?"	Explained using session values and conditional logic to control access.	Used the information when reviewing the role-based menu implementation.
	ChatGPT	Integration testing support	"What should be tested when integrating frontend, backend and database components?"	Suggested checking communication flow, data transfer and error handling.	Used the suggestions to prepare integration testing before prototype submission.
	ChatGPT	Prototype evaluation	"How to write a database implementation section for a software report?"	Suggested including screenshots, working features, repository evidence and testing results.	Used the checklist when preparing materials for Deliverable 3 submission.
Week 13	ChatGPT	Debugging integration problems	"How can I troubleshoot issues between multiple Java Servlet modules?"	Suggested checking request flow, service logic, database updates and component connections.	Used the steps as a guide when investigating integration issues before final evaluation.
	ChatGPT	Final report writing support	"How should architecture implementation be explained in a software project report?"	Provided guidance on explaining architecture decisions, components and system evaluation.	Used the structure as a reference when writing the architecture implementation section of the final report.